

Sourav

Uttarakhand, India • souravsemwall@gmail.com • +91 76682 38542 • Github • Linkedlin

Education

Graphic Era Hill University (GEHU)

2024 – 2028

B.Tech in Computer Science and Engineering

CGPA: 7.5 / 10.0

Technical Skills

Languages	C++, Java, Python, SQL, Bash
Systems	Linux (Fedora), Raw sockets, libpcap, POSIX APIs, TCP/IP internals, Protocol parsing
Backend	FastAPI, REST APIs, JWT, SQLAlchemy, SQLite, PostgreSQL, MySQL
AI / ML	LLMs, RAG, Prompt engineering, AI Agents, Tool-calling, NumPy, Pandas, Scikit-learn
Cryptography	AES-256-GCM, ECDH key exchange, End-to-end encryption
Concepts	Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP
Tools	Git, GitHub, VS Code, Maven, GDB, Valgrind

Projects

Packet Analyzer

C++, Linux, Raw Sockets, libpcap Ongoing

- Engineered a low-level network traffic capture and analysis tool on Linux using raw sockets and **libpcap**, parsing Ethernet, IPv4, TCP, and UDP headers from binary wire data.
- Extracted per-flow statistics (packet counts, byte volumes, protocol distribution) enabling real-time traffic profiling on a live network interface.
- Applied TCP/IP stack internals knowledge to handle fragmented/out-of-order packets; implemented offline analysis via **pcap** trace files.

PeerLink — Serverless Secure P2P File Transfer

Java, JavaFX, TCP/IP, Cryptography 2026

- Built a serverless peer-to-peer file transfer application communicating directly over TCP without a central relay, eliminating single-point-of-failure in file sharing.
- Implemented end-to-end encryption using ECDH for session key agreement and AES-256-GCM for authenticated encryption, providing confidentiality and integrity against MITM attacks.
- Leveraged Java 21 Virtual Threads and a **BlockingQueue** pipeline to concurrently handle multi-file transfers, decoupling encryption from I/O for throughput efficiency.
- Designed an invite-code handshake for peer discovery under a zero-trust network model.

OptiPatrol — Crime Hotspot Prediction & Patrol Optimization

Python, FastAPI, React, SQLite 2026

- Architected a full-stack geospatial crime analysis platform; designed REST API layer with FastAPI exposing endpoints for crime ingestion, hotspot computation, and patrol scheduling.
- Implemented BFS-based geographic clustering and a greedy patrol allocation algorithm over a weighted crime-density graph to generate optimized patrol routes from historical data.
- Resolved N+1 query anti-patterns in the SQLAlchemy ORM schema via eager loading strategies, reducing per-request database round-trips on aggregated crime record queries.

AI Agent Assistant

Python, LLM APIs, RAG, Vector Store 2026

- Built a domain-specific AI assistant using Retrieval-Augmented Generation (RAG), grounding responses in retrieved context to reduce hallucination on specialized document corpora.
- Integrated LLM function-calling API to enable autonomous tool invocation within a conversational agent loop; implemented sliding-window memory and embedding-based semantic retrieval.

Relevant Coursework

Data Structures & Algorithms • Operating Systems • Computer Networks • DBMS • OOP • Discrete Mathematics • Probability & Statistics • Engineering Mathematics

Leadership & Activities

- Technical Club**, GEHU — co-anchored a workshop on transformer architecture and LLMs (“*Attention Is All You Need: How ChatGPT Processes Language*”) for 100+ student attendees.
- Assisted in organizing a 200+ participant university hackathon: managed technical infrastructure, participant onboarding, and multi-track event logistics.